

Omni-LLaMA-AD: A Unified Model for Open-Set Visual Anomaly Detection

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ABSTRACT

Visual anomaly detection (VAD) aims to identify image regions that deviate from established normal patterns. Existing methods often rely on domain-specific training and follow a “one-class-one-model” paradigm, limiting scalability. We propose **Omni-LLaMA-AD**, the first unified multimodal large language model for open-set anomaly detection, capable of handling diverse domains with minimal supervision. Built on a pretrained LLaMA backbone, the model uses a VQGAN-based tokenizer and supports joint vision-language generation. Trained via vision-language alignment and instruction tuning, it achieves effective anomaly detection with only a few normal samples and no domain-specific fine-tuning. Our demo showcases the model’s ability to generate high-quality anomaly masks across industrial, medical, and logical datasets, highlighting its strong cross-domain generalization and interactive dialogue-based user experience.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

Visual Anomaly Detection, Multimodal Representation Learning, Large Language Models, Autoregressive Modeling

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1 INTRODUCTION

Visual anomaly detection (VAD) [1, 2] aims to identify samples that deviate from normal patterns, often indicating critical conditions like defects or pathological changes. However, large domain and

anomaly variability makes existing VAD systems highly specialized—typically requiring separate models per domain or even per object category, following a “one-class-one-model” paradigm. This hinders scalability and limits broader applicability.

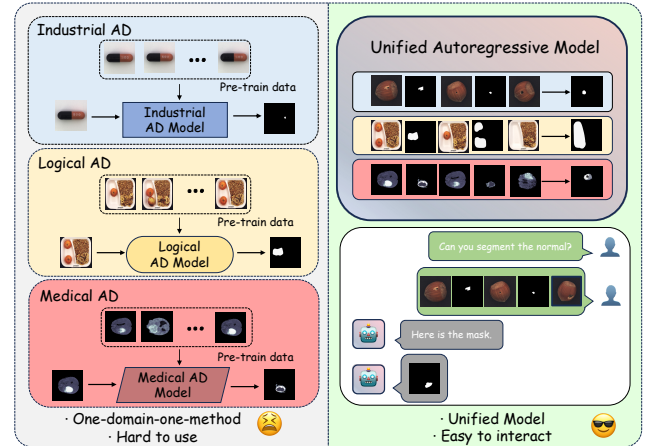


Figure 1: Comparison with Existing VAD Methods. Unlike domain-specific VAD approaches, Omni-LLaMA-AD enables unified, open-set anomaly detection and supports interactive analysis via a chatbot interface.

Recent advances in large language models [3, 4], particularly those based on autoregressive Transformers, have demonstrated remarkable generalization through next-token prediction. Building on this success, recent research has extended LLMs to support image generation, resulting in the emergence of unified multimodal large language models (MLLMs) [5]. These models are capable of processing both language and visual inputs to produce cross-modal outputs.

Inspired by the capabilities of unified MLLMs, we reformulate anomaly detection as a sequence prediction problem. Given an input image sequence and an optional text prompt, the model predicts anomaly masks by learning latent regularities within the sequence. In this framework, the image itself acts as an intrinsic prompt, analogous to text prompts.

In this paper, we present **Omni-LLaMA-AD**, a unified multimodal model for visual anomaly detection that generalizes across

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diverse domains. Unlike conventional approaches, Omni-LLaMA-AD requires no domain-specific training, relying only on a few samples at test time to perform effective anomaly detection, thus greatly enhancing transferability and deployment feasibility.

2 OMNI-LLAMA-AD

2.1 Architecture

Large language models (LLMs) possess strong zero-shot generalization capabilities, enabling them to tackle a wide range of previously unseen tasks. To leverage these capabilities, we propose **Omni-LLaMA-AD**, an end-to-end unified framework for multimodal anomaly detection. As illustrated in Figure 2, Omni-LLaMA-AD comprises a vision encoder and a text encoder to process heterogeneous modalities from user queries Q . Encoded features are concatenated and processed by a pretrained LLaMA model, which autoregressively generates multimodal output representations. These representations are subsequently decoded into either images or text to form the final response R .

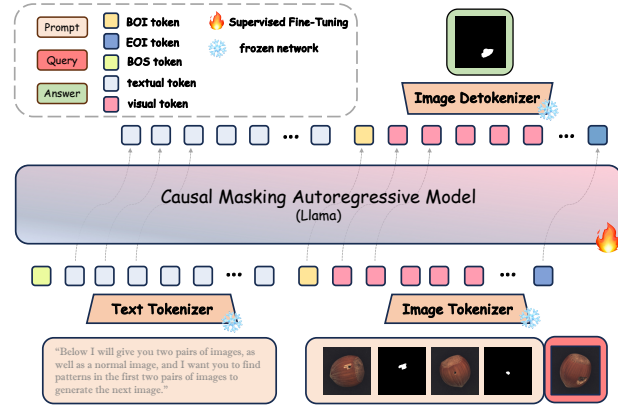


Figure 2: The Architecture of Omni-LLaMA-AD. Omni-LLaMA-AD processes discrete textual tokens and VQ-encoded visual tokens using a decoder-only Transformer for next-token prediction, supporting unified anomaly detection across both language and vision modalities.

Our approach builds upon a decoder-only Transformer architecture, with LLaMA [6] serving as the base model. It models the conditional probability of the t -th token given preceding tokens as $P(x_t | x_1, x_2, \dots, x_{t-1})$ and is trained using standard cross-entropy loss over the token sequence.

2.2 Anomaly Mask Generation

To adapt pretrained LLMs for anomaly detection, we design a two-stage training strategy. In the first stage, we perform vision-language alignment by training the model to generate descriptive texts given an input anomaly image, using a large-scale image-text dataset. In the second stage, we freeze the base LLM and apply instruction tuning with Low-Rank Adaptation[7], enabling task-specific finetuning with minimal trainable parameters.

To enable anomaly mask generation, we reinterpret the task as a constrained image synthesis problem. Specifically, we discretize

images into visual tokens using a pretrained VQGAN [8] model, which compresses input images by a factor of 16. The resulting codebook of 16,384 visual tokens is appended to the LLaMA vocabulary, along with two special tokens, `<start_of_image>` and `<end_of_image>`, to denote visual token sequences. During training, both text and visual tokens contribute to the loss, facilitating multimodal generation.

Through this unified training strategy, Omni-LLaMA-AD is capable of flexibly generating outputs across modalities, including textual descriptions and visual anomaly masks.

2.3 User Experience

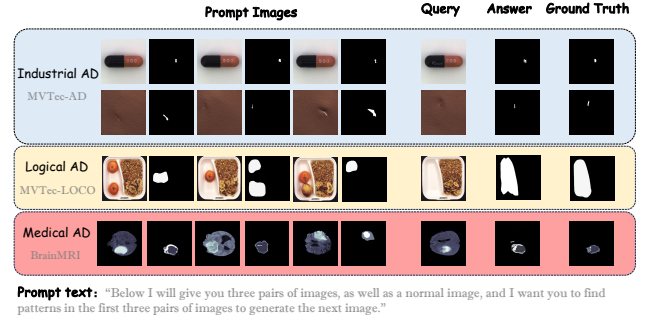


Figure 3: Visualization Results of Omni-LLaMA-AD. Omni-LLaMA-AD accurately segments unseen anomalies across diverse domains using only a few normal references, demonstrating strong few-shot generalization.

We deploy an online interactive demo on a single server equipped with an NVIDIA A100 GPU. The demo provides a conversational interface, allowing users to interact with Omni-LLaMA-AD via text input while uploading multiple images for analysis. We implement streaming decoding to minimize latency, significantly improving user experience. Example results in Figure 3 demonstrate high-quality anomaly mask generation across three diverse datasets: BrainMRI (medical)[9], MVTec LOCO (logical)[10], and MVTec-AD (industrial)[11].

3 CONCLUSION

We introduce **Omni-LLaMA-AD**, a unified multimodal model for open-set visual anomaly detection. Our approach demonstrates the feasibility of leveraging LLMs for anomaly detection tasks with minimal domain-specific training. Extensive experiments and our online demo validate Omni-LLaMA-AD's capacity for broad adaptation, paving the way for practical, scalable deployment of anomaly detection systems.

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